GWCD Documentation - Exceedance Calculations

Vermont AHS GIS and Data Viz | Groundwater Chemistry Database (GWCD)

Exceedance Calculations

Threshold types

Each analyte is checked against up to four benchmark types:

Type	Meaning
MCL	Maximum Contaminant Level (legally enforceable, federal/state drinking water standard)
SMCL	Secondary MCL (aesthetic — taste, odor, staining — not a health standard)
VHA	Vermont Health Advisory
Threshold	A general health-based advisory/action level, used where no MCL/SMCL/VHA applies

A `None` /blank value for a given type means that benchmark doesn't apply to that analyte.

Threshold table (as hardcoded in the script)

Analyte	MCL	SMCL	VHA	Threshold
Arsenic	0.010 mg/L	—	—	0.001 mg/L
Fluoride	4.000 mg/L	2.000 mg/L	—	0.700 mg/L
Gross Alpha	15 pCi/L	—	—	5
Nitrate	10 mg/L	—	5	0
Nitrite	1 mg/L	—	—	0
Total Nitrate and Nitrite	10 mg/L	—	—	5
Sodium	—	250 mg/L	—	15
Chloride	—	250 mg/L	—	50
Manganese	—	0.05 mg/L	0.3 mg/L	0.08 mg/L
Lead	0.015 mg/L	—	0.001 mg/L	—
Iron	—	0.3 mg/L	—	—
Copper	1.3 mg/L	1 mg/L	1.3 mg/L	—
Radon	—	—	4,000 pCi/L	—
Uranium	0.03 mg/L	—	0.02 mg/L	—
Radium	5	—	—	—
TDS	—	500 mg/L	—	—
Turbidity	—	—	—	1
Odor	—	3	—	—
E.coli	0 (presence)	—	—	0 (presence)
Total Coliform	—	—	—	0 (presence)
pH	—	6.5–8.5 (range)	—	—
Hardness	—	—	—	75–151 (range)

> Note: pH's SMCL and Hardness's Threshold are **ranges** — a result outside the range is "above" (flagged), not just a result over a single number.

Result value used for exceedance math (`ResultCalc`)

Unit conversions are applied first:

```
IF AnalyteAlias == "Uranium" AND unit == "ug/L":  numeric_result *= 0.001;   unit = "mg/L"
IF AnalyteAlias == "Uranium" AND unit == "pCi/L": numeric_result *= 0.00149; unit = "mg/L"
IF AnalyteAlias == "Chloride" AND unit == "ug/L": numeric_result *= 0.001;   unit = "mg/L"
```

Then, non-detects are handled by using half the reporting limit (a standard convention), except for a few analytes with special presence/absence logic:

```
IF ResultType == "<" (non-detect) AND AnalyteAlias NOT IN ["Total Coliform", "E.coli", "Hardn
    ResultCalc = ReportingLimit / 2   IF ReportingLimit IS NOT NULL ELSE NULL (excluded from
ELSE:
    ResultCalc = numeric_result
```

Exceedance flag logic (per record, per threshold type)

```
thresholds = LOOKUP(AnalyteAlias, threshold_table)   # (MCL, SMCL, VHA, Threshold)

FOR each threshold_type IN [MCL, SMCL, VHA, Threshold]:
    limit = thresholds[threshold_type]
    IF limit IS NULL:        flag = NULL   # not applicable to this analyte
    ELIF ResultCalc IS NULL: flag = NULL   # can't determine (e.g. non-detect, no reporting l
    ELIF ResultCalc > limit: flag = 1      # Above
    ELSE:                    flag = 0      # At/Below
```

Special cases layered on top of the general logic above:

```
# pH (SMCL is a range):
flag = 1 IF ResultCalc < 6.5 OR ResultCalc > 8.5 ELSE 0

# Hardness (Threshold is a range):
flag = 1 IF ResultCalc < 75 OR ResultCalc > 151 ELSE 0

# E.coli / Total Coliform: if no numeric result, fall back to qualitative result
IF ResultCalc IS NULL:
    flag = 1 IF WHResultValueAlpha IN ["Present", "Detected"]
    flag = 0 IF WHResultValueAlpha IN ["Absent", "Not detected"]
    flag = NULL otherwise
```

Town/analyte aggregation (public map + summary tables)

```
GROUP deduplicated records BY (AnalyteAlias, TOWNNAME)

FOR each group:
    CityCount = COUNT(records with a non-null Result)
    FOR each threshold_type:
        CityAbove[type] = SUM(flag == 1)
        PctAbove[type]  = CityAbove[type] / CityCount * 100
    Mean, Median, Min, Max, 95th Percentile, StdDev = stats(ResultCalc) for the group
    # rounding precision varies by analyte (e.g. Arsenic/Nitrate to 4 decimals,
    # Fluoride to 3, Gross Alpha/Radon to 1, Uranium to 6)

SUPPRESS group from public output IF CityCount < 20
```